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Project Proposal

Problem Statement & Motivation: Young professionals and college students want to save money while still enjoying “nice” products, makeup, skincare, clothing, tech accessories, decor, and more. They struggle with discovery friction (it’s hard to find trustworthy dupes or cheaper alternatives across many sites), price–value confusion where they are unsure when a premium product is “worth it” vs. when a dupe is good enough, and the time cost of manually comparing ingredients, materials, reviews, and prices is tedious. The primary target users are budget-conscious college students and early-career professionals who shop online and care about aesthetics/quality.

The reason why an agentic AI solution would be appropriate for this problem is that finding dupes often requires chaining tasks, understanding the original product, searching across categories/sites, comparing attributes, and balancing price vs. quality (dynamic, multi-step reasoning). Also, the continuous updating of prices, availability, and trends changes frequently. Agents can re-check, re-rank, and adjust recommendations over time.

Use Cases & Scenarios: In the first scenario, a user wants a cheaper dupe for a specific product, perhaps a \$50 foundation or a \$200 pair of headphones. The system begins by analyzing the

original product, extracting key attributes such as category, features, ingredients, or technical specifications. A dedicated discovery agent then searches across retailers for similar items, while a scoring agent evaluates each candidate based on similarity, price savings, and overall value. The final output is a curated list of dupes accompanied by clear explanations of why each one is recommended.

A second scenario centers on ongoing price monitoring. A user may be willing to wait for a sale or a price drop before purchasing a product or its dupe. The system maintains a watchlist, periodically checks prices, and notifies the user when a product meets their target price or when a new, better-value dupe becomes available. This continuous monitoring is something that would be impractical for a human to perform manually.

Initial System Design: The system incorporates a multi-agent architecture. An orchestrator agent manages the overall workflow and interacts directly with the user. A product understanding agent parses product descriptions and extracts structured attributes. A dupe discovery agent searches across retailers and marketplaces. A similarity and value scoring agent evaluates alternatives. A budget optimization agent handles multi-item wishlists. A price monitoring agent tracks changes over time. Finally, a critic agent checks for hallucinations or unsupported claims, ensuring that recommendations remain grounded in real data, and a summarization agent translates internal reasoning into clear, user-friendly explanations.

The system relies on a variety of tools and data sources, including retailer search functions, product pages, and price comparison services. Users may provide product links, names, or

wishlists directly. Lightweight storage solutions, such as Google Sheets or Notion, can effectively maintain watchlists and user preferences. While scraping may be limited by terms of service, the system can fall back on user-provided data or general search queries when necessary.

Feasibility & Risks: Hallucination is a significant risk, particularly when the AI attempts to infer product attributes that are not explicitly stated. To mitigate this, the critic agent flags uncertain information and prompts the user to verify details. Data access is another constraint, as many retailers restrict automated queries. Context length and computational cost also pose challenges when processing large wishlists or numerous candidate dupes.

The execution timeline for this project is structured around the course's required milestones. The project proposal is due on February 4, establishing the foundation of the problem, user needs, and system architecture. By March 11, I will complete Project Milestone I, which focuses on building a simple front-end interface, either a web or mobile interface, that allows users to input product links, wishlists, or budget constraints. Project Milestone II, due on March 25, requires the implementation of the multi-agent workflow itself. At this stage, the orchestrator, product understanding agent, dupe discovery agent, and scoring agent will work together to produce end-to-end dupe recommendations. Project Milestone III, due on April 1, centers on evaluation, gathering user feedback, measuring relevance and accuracy, and assessing the clarity of explanations. The final presentation on April 16 will showcase the system's capabilities, reflect on its limitations, and outline opportunities for future improvement.